produced in gold, as expressed by the formula:

$$_{79}Au^{197} + _{0}n^{1} \rightarrow _{79}Au^{198} \rightarrow _{80}Hg^{198} + _{-1}e^{0}$$

which indicates the emission of an electron, it will produce only one of seven stable mercury isotopes, six of which occur in ordinary mercury in practically equal proportions. This mercury isotope is very suitable for certain optical researches. Namely, when ordinary mercury vapour—in other words, the natural mixture of mercury isotopes—is caused to glow by an electric charge, the spectra of the individual isotopes vary but very slightly, and their spectral lines become superimposed, in what we call a *fine structure*. This fine structure is absent in the mercury produced from gold, and therefore this particular mercury isotope is most suitable for standard spectroscopic measurements (as was proposed some years ago by W. E. Williams), where the essential aim is to obtain lines as sharply defined as possible. If we recall the purpose mentioned at the beginning of this chapter, the production of more valuable substances out of those of lesser value, we shall appreciate that it is an interesting indication of the transmutability of all things that in this particular instance mercury is more valuable than gold, instead of vice versa.

This concludes our survey of the applications of nuclear physics. All that we have discussed represents merely the beginnings of a development, the future progress of which cannot even be estimated. But practical applications are not the most important aspect of nuclear physics, and that is why they have not been discussed in greater detail in these lectures; the practical benefits of a knowledge of natural phenomena constitute a later problem. For the time being, the main thing is to understand the structure of the nucleus of the atom. The purpose of these lectures has been to present a summary of what has been accomplished in this field and what still remains to be done. All that has been described here was intended to impart both to listeners and readers something of the magical effect on all of us of those natural phenomena which are so difficult of access, and in particular of those whose internal laws we have not yet fathomed.